# Adamya Verma

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.68 |
| Shri Ram Public School | Class XII (CBSE) | 2023 | 94.40% |
| Shri Ram Public School | Class X (CBSE) | 2021 | 94.80% |

## WORK EXPERIENCE

**Oracle | Software Engineering Intern**

*OptQuest - Automated SQL Optimizer Bug Detection Framework*

- Discovered and reported **1 wrong-result bug** and **1 database crash** in the **Oracle Database** optimizer through automated differential testing.
- Designed and implemented **61 SQL transformation families** spanning **10+ Oracle optimizer transformations** (**Subquery Unnesting**, **Materialized View Rewrite**, **Predicate Pushdown**, **OR Expansion**, **View Merging**).
- Developed **OptQuest**, an automated SQL differential testing framework in **Python** that generates semantically equivalent SQL variants using optimizer hints to exercise diverse optimizer code paths.
- Built an automated validation pipeline comparing query results, execution plans, and optimizer traces on deterministic edge-case datasets to identify optimizer defects.

## PROJECTS

**Lightweight Countermeasure for Shared LLC Attacks Using Cache Partitioning** *(Team of 2)*

- Designed and implemented **UCP-RI** and **UCP-CI**, two lightweight extensions to **Utility-Based Cache Partitioning (UCP)** that invalidate stale cache blocks during repartitioning to mitigate shared LLC side-channel attacks.
- Evaluated across **50** multi-programmed **SPEC CPU2006/2017** and graph workloads in **ChampSim**, achieving up to **3.3% weighted speedup** over **LRU** while preserving cache isolation between co-located processes.

**RISC-V Simulator with Cache Simulation** (Team project - 2 members)

- A custom simulator supporting step-wise execution and configurable direct-mapped, set-associative, and fully-associative cache.
- Implemented cache mapping logic and actively participated in system testing and debugging.

**Virtual Memory Management in gemOS** (Team project - 2 members)

- Extended gemOS to simulate Linux-style memory behavior using mmap, munmap, and Copy-on-Write mechanisms.
- Implemented mmap, page fault handling, and Copy-on-Write logic during fork.

**Meeting and Communication Management System** (Team of 6)

- Engineered a full-stack web application using React, Node.js, WebSockets, PostgreSQL, and the Jitsi API to facilitate real-time video conferencing, synchronized agenda tracking, and collaborative task management.
- Integrated a Python-based AI intelligence layer utilizing the Sarvam AI SDK and Large Language Models (LLMs) for bilingual audio transcription, real-time meeting summarization, and automated action-item extraction.

**GraphX: A Domain-Specific Language for Graph Processing** (Team of 7)

- Built the compiler frontend for **GraphX**, a domain-specific language for graph processing -- implemented **lexical analysis**, **parsing**, and **semantic validation** for a custom syntax supporting dynamic graph manipulation.
- Designed grammar and semantic rules to natively express **graph algebra** operations (**tensor products**, **metric closures**), enabling the backend team to implement density-aware memory optimizations without additional syntax-level changes.

## SKILLS

- **Languages:** C++,C,Python,JS,HTML,CSS
- **Python Libraries:** Numpy,Pandas,Scikit-learn,Pytorch,Tensorflow,Matplotlib
- **Database Skills:** SQL, PostgreSQL, MongoDB (NoSQL)
- **Dev Tools:** Git,Docker,VS Code,Bash,LaTeX,npm
- **Web Development:** React, Next.js, HTML, CSS, JavaScript

## SCHOLASTIC ACHIEVEMENTS

- **Gold Rank** - Flipr Labs GenAI Hackathon
- Achieved a rating of **1400 (Specialist)** on **Codeforces**; **3-Star Coder** on **CodeChef**
- **Silver Medal**, Inter IIT Tech Meet 14.0 -- **Genuity: Algorithmic Optimisation**

## RELEVANT COURSES

- **Computer Science:** Operating System, Computer Architecture, Computer Networks, Distributed Computing, Compilers, Foundations of Machine Learning
- **AI:** Deep Learning, Natural Language Processing
- **Mathematics:** Linear Algebra,Probability and Random Variables,Intro to Statistics,Transform Techniques,calculus

## POSITIONS OF RESPONSIBILITY

- Teaching Assistant, **CS3563 (Introduction to DBMS II)** and **CS2323 (Computer Architecture)**, IIT Hyderabad
- **Head, ML Domain**, Tinkerers' Lab -- led AI/ML workshops and hackathons for student engagement

## EXTRACURRICULAR

- **Represented IIT Hyderabad** in **Inter IIT Tech Meet 14.0** - Problem Statements: **Observe.AI - Causal Rationale Extraction and Synthesis from Conversational Data on Business Events** and **Genuity - Algorithmic Optimisation**.